Do Now · retrieval

Software development methods: evaluation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Independent tasks take 3 and 6 min; final integration 2 min. What is ideal parallel duration?

Q2

Name one XP practice intended to find code errors early.

Q3

Why would a user evaluate a prototype?

Q4

Simplify F=(¬A ∧ ¬B) ∨ (¬A ∧ B). State the law and final answer.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Avoid a universal answer — Method suitability depends on risk, requirements, regulation, user access and timescale.

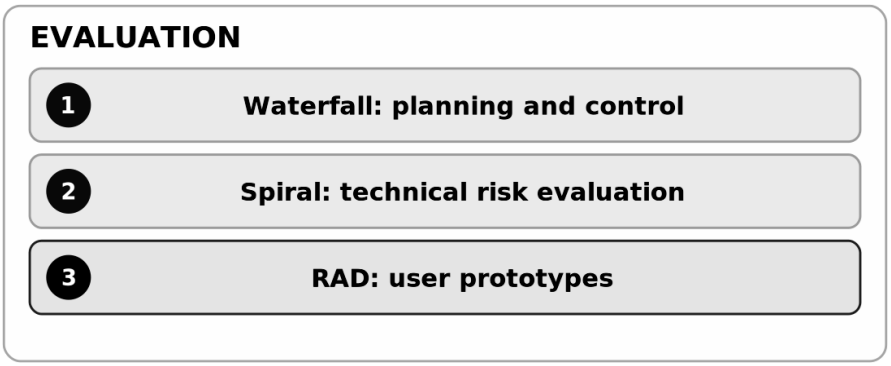
Core idea 2

Support with scenario evidence — A high-risk system may justify Spiral's explicit risk analysis despite extra overhead.

Core idea 3

Include a drawback — Evaluate cost, documentation, flexibility and technical quality rather than merely naming a method.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

evaluation — Judgement supported by evidence and limitations.

risk-driven — Development priorities guided by significant uncertainties.

iterative — Revisits improvements in cycles.

stakeholder — Person or organisation affected by a project.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A safety-critical device uses unfamiliar hardware. Evaluate Spiral versus RAD.

Follow the worked method

1. Predict the method — Recommend Spiral because technical risks can be evaluated at each iteration.

2. Check and explain — Explain the disadvantage: risk analysis adds project management and cost.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the method
RAD fits frequent volunteer feedback and a requirement to develop the interface rapidly.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Give one defining feature each of Waterfall, Spiral, RAD and XP.

Question 2

Choose a development method for a regulated safety-critical control application with uncertain technology. Give three reasons and one drawback.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate Agile compared with Waterfall for a government reporting system where requirements are partly fixed but user interface needs may change.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The best development methodology is the same for every software project.

Use key vocabulary from memory

1. Explain evaluation in your own words.

2. Explain risk-driven in your own words.

3. Explain iterative in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.